1. $G, R.$ $\mathrm{tr}[T_R^a T_R^b] = T(R)\,\delta^{ab}$

$D(R)$ $T_R^a T_R^b = C_2(R)\,\mathbb{1}$

$SU(N)$: $R_F$ $D(R_F) = N,$ $T(R_F) = \frac{1}{2},$ $C_2(R) = \frac{N^2-1}{2N}.$

$R_A$ $D(R_A) = N^2-1$

$SO(N)$: $R_F$ $D(R_F) = N,$ $T(R_F) = 2.$ $C_2(R_F) = N-1$

$R_A$ $D(R_A) = \frac{N(N-1)}{2}$

$T(R)\,D(R_A) = C_2(R)\,D(R).$

2. $SU(N)$, $SO(N)$ $R_F$.

- 无迹厄米矩阵
- 实反对称矩阵

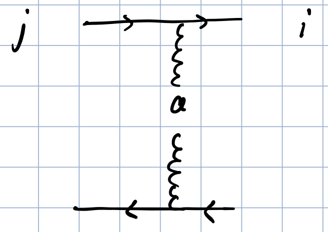
} 的独立正交基.

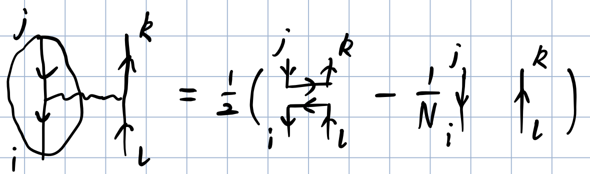
3. Fierz identity

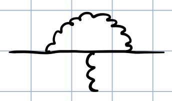
$SU(N)$, $R_F$. $(T_F^a)^i{}_j (T_F^a)^k{}_l = \frac{1}{2}\left(\delta^i_l\,\delta^k{}_j - \frac{1}{N}\delta^i_j\,\delta^k_l\right)$

$\delta^i{}_j$ : $j \longrightarrow i$ $\delta^{ab}$ : $a \;\sim\!\sim\!\sim\; b$

$(T_F)^i{}_j$ :







4. 表示的直和与直积

5. 直积分解.

$$\bar{R} \otimes R = 1 \oplus \cdots\cdots$$

$$\begin{tikzcd} \bar{R} \arrow[d] & R \arrow[d] \\ \chi^{\dagger i} \times & \psi_j \end{tikzcd}$$

$$\overset{\shortparallel}{\underbrace{\Phi^i{}_j}} = \frac{\underbrace{\delta^i{}_j}_{\begin{pmatrix}1&&\\&\ddots&\\&&\end{pmatrix}}}{N}\,\mathrm{tr}(\Phi) + \underbrace{\tilde{\Phi}^i{}_j}_{\text{无迹.}}$$

两边求 trace.

$$\mathrm{tr}(\Phi) = \mathrm{tr}(\Phi)\cancel{\frac{N}{N}} + \underbrace{\mathrm{tr}(\tilde{\Phi})}_{}=0$$